

ID3 decision tree example using the weather/golf dataset:

Dataset: Weather Conditions & Golf Play

Day	Outlook	Temperature	Humidity	Wind	Play Golf
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

14	Rain	Mild	High	Strong	No
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Key:

Target: Play Golf (Yes/No)

Features: Outlook, Temperature, Humidity, Wind

Step 1: Calculate Root Node

Entropy of entire dataset (S):

14 samples: 9 "Yes", 5 "No"

$$H(S) = -[(9/14)\log_2(9/14) + (5/14)\log_2(5/14)] = 0.940$$

Information Gain for each feature:

Feature	IG(S, Feature)
Outlook	0.246 (Highest)
Humidity	0.151
Wind	0.048
Temperature	0.029

Decision: Outlook becomes root node (highest gain).

Step 2: Split by Outlook

Branch 1: Sunny (5 samples)

Samples: Days 1, 2, 8, 9, 11

Entropy: $H(\text{Sunny}) = -[(2/5)\log_2(2/5) + (3/5)\log_2(3/5)] = 0.971$

New gains:

Feature	IG(Sunny, Feature)
Humidity	0.971 (Highest)
Wind	0.020
Temperature	0.571

Split by Humidity:

High (Days 1,2,8) → **No** (pure leaf)

Normal (Days 9,11) → **Yes** (pure leaf)

Branch 2: Overcast (4 samples)

Samples: Days 3,7,12,13 → All "Yes"

Leaf Node: Yes

Branch 3: Rain (5 samples)

Samples: Days 4,5,6,10,14

Entropy: $H(\text{Rain}) = -[(3/5)\log_2(3/5) + (2/5)\log_2(2/5)] = 0.971$

New gains:

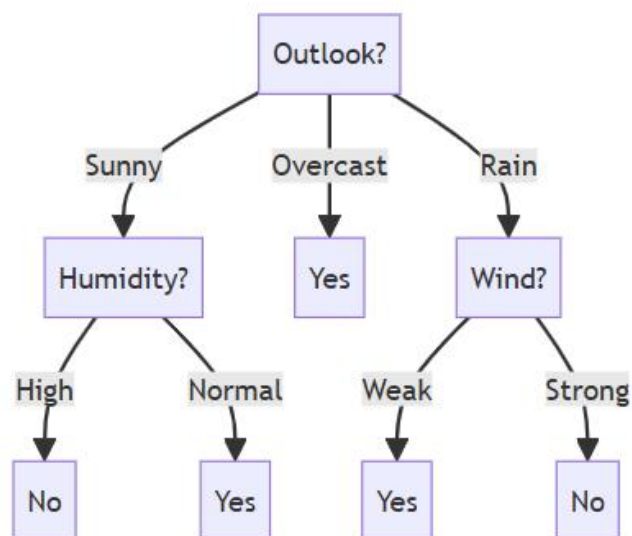
Feature	IG(Rain, Feature)
Wind	0.971 (Highest)
Humidity	0.020
Temperature	0.020

Split by Wind:

Weak (Days 4,5,10) → Yes

Strong (Days 6,14) → No

Final Decision Tree



Tree Interpretation:

1. Check **Outlook**:

● If *Overcast* → **Play Golf = Yes**

● If *Sunny* → Check **Humidity**:

● *High* → **No**

● *Normal* → **Yes**

● If *Rain* → Check **Wind**:

● *Weak* → **Yes**

● *Strong* → **No**

This tree perfectly classifies all 14 samples using the ID3 algorithm's entropy and information gain calculations.